

1. **Split the data into 88 train segments and 22 test segments.** Segments are determined when the time increases by more than 20ms
2. **Compare the cross-validated area under curve (CV AUC) for every predictor**

Top predictors are shown in the image below:

```
Top single predictors (CV AUC):  
  
           feature    cv_auc  
Motor current(mA)__min 0.950984  
Motor current(mA)__mean 0.808407  
Motor electrodynamic force__std 0.799258  
Motor electrodynamic force__max 0.761354  
Motor electrodynamic force__mean 0.751604  
  
Best single predictor: Motor current(mA)__min
```

As **minimum Motor current** (in a segment) has the highest CV AUC, we will include it as the first predictor in the model.

3. Compare the CV AUC for other predictors

Top predictors are shown in the image below:

```
Top 2nd predictors to add (CV AUC):  
  
           feature    cv_auc  
Motor current(mA)__mean 1.000000  
Motor Voltage(10mV)__mean 1.000000  
Motor Voltage(10mV)__max 0.999615  
Motor Voltage(10mV)__std 0.985582  
Door leaf position__std 0.967522  
Best 2nd predictor: Motor current(mA)__mean
```

As **mean Motor current** (in a segment) has the highest CV AUC, we will include it as the first predictor in the model.

Since the CV AUC is 1, there is no need to add more unnecessary predictors.

4. Use logistic regression model to determine the Coefficients of the model

```
INTERCEPT = -30.297485946310392  
B_CUR_MIN = 0.21513995728309812  
B_CUR_MEAN = 0.03870289025007016
```